

```
import cv2
import matplotlib.pyplot as plt

# Load image (replace with your autonomous driving image)
image = cv2.imread("road.jpg")

if image is None:
    print("Image not found!")
    exit()

# Convert BGR to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Image Formation Model Improvements

# 1. Noise Reduction
denoised = cv2.GaussianBlur(image_rgb, (5, 5), 0)

# 2. Contrast Enhancement
gray = cv2.cvtColor(denoised, cv2.COLOR_RGB2GRAY)
equalized = cv2.equalizeHist(gray)

# 3. Edge Detection
edges = cv2.Canny(equalized, 100, 200)

# Display Results
plt.figure(figsize=(12,8))

plt.subplot(2,2,1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(denoised)
plt.title("Noise Reduced")
plt.axis("off")
```

```
plt.subplot(2,2,3)
plt.imshow(equalized, cmap="gray")
plt.title("Contrast Enhanced")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(edges, cmap="gray")
plt.title("Edge Detection")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Image not found!

error Traceback (most recent call last)
[/tmp/ipykernel_18208/603449614.py](#) in <cell line: 0>()

```
10
11 # Convert BGR to RGB
---> 12 image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
13
14 # Image Formation Model Improvements
```

error: OpenCV(4.13.0) /io/opencv/modules/imgproc/src/color.cpp:199: error: (-215:Assertion failed) !_src.empty() in function 'cvtColor'

Next steps: [Explain error](#)

